THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · HARD LEVEL CHAPTER TEST SET 3

Chapter 6: **Control and Coordination** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A researcher removes the pituitary of a young rat. Expected result: 1

(a) gigantism **(b)** faster growth **(c)** no change **(d)** stunted growth

2. Assertion (A): Auxin promotes cell elongation on the shaded side of a shoot. Reason (R): Light causes auxin to diffuse to the shaded side. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

3. A potted plant is kept lying on its side. After some days the tip of its shoot bends upward. This is an example of: 1

(a) negative geotropism **(b)** positive phototropism
(c) hydrotropism **(d)** positive geotropism

4. A tall person has overactive: 1

(a) pituitary **(b)** pancreas **(c)** adrenal **(d)** thyroid

5. Assertion (A): Adrenaline is called the emergency hormone. Reason (R): It prepares the body to deal with stress by increasing heart rate. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

6. Which of the following is correct for hormones in plants and animals? 1

(a) Both are chemical messengers **(b)** Both travel in blood
(c) Both are secreted by glands **(d)** Both are nerve impulses

7. An organism without a thyroid will lack: 1

(a) testosterone **(b)** insulin **(c)** adrenaline **(d)** thyroxine

8. Assertion (A): A person cannot walk in a straight line after cerebellum injury. Reason (R): Cerebellum controls balance and posture. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

9. **Assertion (A):** Shoots of a plant bend towards light. 1
Reason (R): Auxin accumulates on the shaded side of the shoot and makes the cells there grow longer.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** The pituitary gland is called the master gland. 1
Reason (R): The pituitary secretes growth hormone, which regulates growth and development of the body.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Explain why growth hormone injections can help a short child but only if given early. 2

12. Name the hormone present in higher concentration in the shaded side of a shoot and its effect on the shoot and the root. 2

13. Why does an insulin injection need to be given to a person with type 1 diabetes and not oral tablets of insulin? 2

14. Why do tall plants stay upright against gravity? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. How would a plant respond in a space shuttle without gravity? Explain. 3

16. Why is the loss of the cerebellum more serious for a dancer? 3

17. What is the importance of the myelin sheath? (basic idea) 3

OR

Explain how the sensitive plant protects itself by the movement of its leaves. 3

18. Explain why thyroxine deficiency in infants may slow down growth of brain. 3

OR

How does a reflex arc differ in plant and animal responses? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Sunita kept a potted seedling near a window. After a few days she found the shoot bent towards the window. She also noticed that the tendrils of her pea plant coiled round a support, and the roots of the seedling grew downwards. 5
(i) Name the movement of the shoot towards light.
(ii) Name the plant hormone responsible for this bending.
(iii) On which side of the shoot does this hormone accumulate?
(iv) Name the movement shown by the tendril and explain how it occurs.

20. Explain the possible effects on the body if a person's hindbrain gets damaged in an accident. 5

OR

Discuss how a sudden stimulus like a loud noise leads to a coordinated response of the nervous and endocrine system. 5

END OF QUESTION PAPER — Total: 40 Marks